

Marital Sorting using Regressions

Bastian Schulz¹, Jakob Sørensen², and Jesper R.-V. Sørensen³

Nordic Data Meeting - April 2026

¹Aarhus University, IZA & CESifo

²University of Copenhagen, CEPR & CESifo

³University of Copenhagen & Aarhus Center for Econometrics (ACE)

Introduction: Marital Sorting

The sorting question: who marries whom, and why?

→ central question in economics, sociology, and demography (Becker, 1973, 1974, 1981)

Introduction: Marital Sorting

The sorting question: who marries whom, and why?

→ central question in economics, sociology, and demography (Becker, 1973, 1974, 1981)

Marital sorting has been shown to

- affect educational choice, family formation, labor supply, investments in human capital and children, and earnings dynamics (Chiappori et al., 2017; Goussé et al., 2017; Chiappori et al., 2018; Kirkeboen et al., 2022; Almar et al., 2025a)

Introduction: Marital Sorting

The sorting question: who marries whom, and why?

→ central question in economics, sociology, and demography (Becker, 1973, 1974, 1981)

Marital sorting has been shown to

- affect educational choice, family formation, labor supply, investments in human capital and children, and earnings dynamics (Chiappori et al., 2017; Goussé et al., 2017; Chiappori et al., 2018; Kirkeboen et al., 2022; Almar et al., 2025a)
- dampen or amplify income inequality within and across households (Greenwood et al., 2014, 2016; Adda et al., 2017; Eika et al., 2019; Santos and Tertilt, 2024; Almar et al., 2025b; Goussé et al., 2025)

Introduction: Marital Sorting

The sorting question: who marries whom, and why?

→ central question in economics, sociology, and demography (Becker, 1973, 1974, 1981)

Marital sorting has been shown to

- affect educational choice, family formation, labor supply, investments in human capital and children, and earnings dynamics (Chiappori et al., 2017; Goussé et al., 2017; Chiappori et al., 2018; Kirkeboen et al., 2022; Almar et al., 2025a)
- dampen or amplify income inequality within and across households (Greenwood et al., 2014, 2016; Adda et al., 2017; Eika et al., 2019; Santos and Tertilt, 2024; Almar et al., 2025b; Goussé et al., 2025)

→ **Bottom line:** who marries whom is important to understand household decisions at the micro level and their implications at the macro level.

State of the art in the sorting literature

How to measure sorting? A variety of statistical and model-based methods is available. Key empirical regularities have been documented in the literature ...

State of the art in the sorting literature

How to measure sorting? A variety of statistical and model-based methods is available. Key empirical regularities have been documented in the literature ...

BUT:

- focus on *one-dimensional* matching, specifically matching based on *education*

State of the art in the sorting literature

How to measure sorting? A variety of statistical and model-based methods is available. Key empirical regularities have been documented in the literature ...

BUT:

- focus on *one-dimensional* matching, specifically matching based on *education*
- focus on *same-type couples* (i.e., positive assortative matching, PAM);

State of the art in the sorting literature

How to measure sorting? A variety of statistical and model-based methods is available. Key empirical regularities have been documented in the literature ...

BUT:

- focus on *one-dimensional* matching, specifically matching based on *education*
- focus on *same-type couples* (i.e., positive assortative matching, PAM);
- (i) quite specific statistical techniques (e.g., likelihood ratios, odds ratios, correlations) and/or (ii) structural matching models with strong assumptions (e.g., frictionless matching, preference shocks, parametric “home production” functions)

State of the art in the sorting literature

How to measure sorting? A variety of statistical and model-based methods is available. Key empirical regularities have been documented in the literature ...

BUT:

- focus on *one-dimensional* matching, specifically matching based on *education*
- focus on *same-type couples* (i.e., positive assortative matching, PAM);
- (i) quite specific statistical techniques (e.g., likelihood ratios, odds ratios, correlations) and/or (ii) structural matching models with strong assumptions (e.g., frictionless matching, preference shocks, parametric “home production” functions)

→ **Bottom line:** literature is relatively narrow in scope, methodologically scattered, and comparability across studies is limited. Affects applicability in the wider applied literature.

See examples from applied literature

Contributions

We develop new *regression-based* tools that are designed to make it easy for the applied economist to study marital sorting directly or use it as an outcome in other settings.

We develop new *regression-based* tools that are designed to make it easy for the applied economist to study marital sorting directly or use it as an outcome in other settings.

1. **Formalization:** Develop a general and multidimensional framework to measure sorting.
 - Core estimand: the selection kernel; generalizes a well-known measurement strategy: the likelihood that a couple forms relative to random matching
 - Flexible, but potentially (very) high-dimensional object. Computational challenges.
 - Key question: how to calculate random matching probabilities efficiently? Simulation.

We develop new *regression-based* tools that are designed to make it easy for the applied economist to study marital sorting directly or use it as an outcome in other settings.

1. **Formalization:** Develop a general and multidimensional framework to measure sorting.
 - Core estimand: the selection kernel; generalizes a well-known measurement strategy: the likelihood that a couple forms relative to random matching
 - Flexible, but potentially (very) high-dimensional object. Computational challenges.
 - Key question: how to calculate random matching probabilities efficiently? Simulation.
 - Collapsed kernel: often, we are interested in low-dimensional (but non-linear) functions defined on the type-space of individual- and couple-level covariates
 - e.g., same education, male breadwinner, older wife, taller husband, etc.
 - In simple cases, the collapsed kernel is just a dummy.
 - Reduces dimensionality, but is not restricted to “same-type”.

2. **Implementation:** Combine regressions with simulation:

- We show that (collapsed) selection kernels can be computed/approximated using regressions. Two approaches are available (asymptotically equivalent).
 - Population-based: directly infer random matching probabilities from the data
 - Simulation-based: draw random couples from marginal type distributions

2. **Implementation:** Combine regressions with simulation:

- We show that (collapsed) selection kernels can be computed/approximated using regressions. Two approaches are available (asymptotically equivalent).
 - Population-based: directly infer random matching probabilities from the data
 - Simulation-based: draw random couples from marginal type distributions
- Advantage of the regression approach:
 - Straightforward to implement, multidimensionality and complex sorting patterns through “controls” (discrete/continuous, parametric/non-parametric), no structural model
 - Use std. language and tools from applied microeconometrics, integrate into quasi-experimental settings when the composition of the population is potentially affected.

2. **Implementation:** Combine regressions with simulation:

- We show that (collapsed) selection kernels can be computed/approximated using regressions. Two approaches are available (asymptotically equivalent).
 - Population-based: directly infer random matching probabilities from the data
 - Simulation-based: draw random couples from marginal type distributions
- Advantage of the regression approach:
 - Straightforward to implement, multidimensionality and complex sorting patterns through “controls” (discrete/continuous, parametric/non-parametric), no structural model
 - Use std. language and tools from applied microeconometrics, integrate into quasi-experimental settings when the composition of the population is potentially affected.
- Disadvantage:
 - Potential simulation and approximation errors.
 - Causality, but we can produce “counterfactuals” by shutting down sorting channels.

2. **Implementation:** Combine regressions with simulation:

- We show that (collapsed) selection kernels can be computed/approximated using regressions. Two approaches are available (asymptotically equivalent).
 - Population-based: directly infer random matching probabilities from the data
 - Simulation-based: draw random couples from marginal type distributions
- Advantage of the regression approach:
 - Straightforward to implement, multidimensionality and complex sorting patterns through “controls” (discrete/continuous, parametric/non-parametric), no structural model
 - Use std. language and tools from applied microeconometrics, integrate into quasi-experimental settings when the composition of the population is potentially affected.
- Disadvantage:
 - Potential simulation and approximation errors.
 - Causality, but we can produce “counterfactuals” by shutting down sorting channels.

3. **Applications:** Revisit old and explore some new questions...

Example

The data slide

Standard full population administrative data from Denmark

Focus on heterosexual couples (married and cohabiting) in one calendar year (2015).

We observe 1,085,788 couples.

Possible Extensions:

- Compare estimates over time
- Include singles (extensive margin)
- Allow for homosexual couples (treating sex as any other covariate).
- Add new data sources (e.g., physical characteristics)

Example: age-based sorting

Table 1: Contingency table of Danish couples by age group (above/below 45) in 2015

		Men		
		Young	Old	All
Women	Young	40.30	7.66	47.96
	Old	1.87	50.18	52.04
	All	42.16	57.84	100.00

The **likelihood ratio** or *selection intensity* captures the excess likelihood (odds) of a particular type combination relative to random matching:

$$S(t, t') = \frac{\Pr(W = t, M = t')}{\Pr(W = t) \Pr(M = t')} \quad (1)$$

For example, $S(\text{Young}, \text{Young}) = 40.30\% / (42.16\% \cdot 47.96\%) \approx 2$

Collapsed Selection Kernel

Rather than examining all four cells separately, we often want to answer simpler “aggregate” questions: e.g., *do couples sort by age group?*

We *collapse* the selection kernel by defining a couple-level characteristic X :

- Let $X = 1$ if both partners are in the same age group (young-young or old-old)
- Let $X = 0$ if partners are in different age groups

Collapsed Selection Kernel

Rather than examining all four cells separately, we often want to answer simpler “aggregate” questions: e.g., *do couples sort by age group?*

We *collapse* the selection kernel by defining a couple-level characteristic X :

- Let $X = 1$ if both partners are in the same age group (young-young or old-old)
- Let $X = 0$ if partners are in different age groups

We aggregate the main-diagonal and off-diagonal likelihood ratios by computing the weighted average, where weights are the expected frequencies under random matching:

$$S_X(1) = \frac{50.18\% + 40.30\%}{57.84\% \times 52.04\% + 42.16\% \times 47.96\%} = 1.798$$

$$S_X(0) = \frac{7.66\% + 1.87\%}{57.84\% \times 47.96\% + 42.16\% \times 52.04\%} = 0.192$$

Same-age couples occur at 179.8%, different-age couples at only 19.2% of random matching

Link to Logistic Regression

The odds ratio comparing same-age to different-age couples is:

$$\mathcal{O}_X = \frac{S_X(1)}{S_X(0)} = \frac{1.798}{0.192} = 9.375$$

The odds ratio comparing same-age to different-age couples is:

$$\mathcal{O}_X = \frac{S_X(1)}{S_X(0)} = \frac{1.798}{0.192} = 9.375$$

This can be recovered through a simple logistic regression:

$$\Pr(Y = 1|D) = \Lambda(\beta_0 + \beta_1 D)$$

where $Y = 1$ for realized couples, $Y = 0$ for random couples, and $D = 1$ for same-age group couples.

The odds ratio comparing same-age to different-age couples is:

$$\mathcal{O}_X = \frac{S_X(1)}{S_X(0)} = \frac{1.798}{0.192} = 9.375$$

This can be recovered through a simple logistic regression:

$$\Pr(Y = 1|D) = \Lambda(\beta_0 + \beta_1 D)$$

where $Y = 1$ for realized couples, $Y = 0$ for random couples, and $D = 1$ for same-age group couples.

The regression yields:

- $\exp(\beta_1) = 9.375$ (the odds ratio)
- $\exp(\beta_0) = 0.192$ (baseline selection intensity for different-age couples)
- $S_X(1) = \exp(\beta_0) \times \exp(\beta_1) = 0.192 \times 9.375 = 1.798$

Three Approaches to Calculate Selection Intensities

	(a)	(b)	(c)	(d)	(e)
	Population-Based		Simulation-Based		
	Exact (LR)	Regression (Poisson)	Regression (Logistic), Simulated Couples (M)		
			108,578	1,085,788	10,857,880
$\exp(\beta_1)$ (Same Age)	9.375	9.375	9.280 (0.064)	9.333 (0.035)	9.365 (0.031)
$\exp(\beta_0)$ (Constant)	0.192	0.192	0.193 (0.001)	0.192 (0.001)	0.192 (0.001)
$S_X(1)$ (Likelihood ratio)	1.798	1.798 $= \exp(\beta_0) \times \exp(\beta_1)$	1.791	1.792	1.798
Realized couples (N)		1,085,788			

The simulation-based approach recovers the exact values precisely, even with relatively few simulated couples.

With the simple age-based example, we have demonstrated:

- How to compute selection kernels (likelihood ratios) from contingency tables
- How to collapse the kernel into aggregate measures (same vs. different age)
- How logistic regression recovers these measures from stacked data
- That simulation-based estimation is accurate even with modest sample sizes

Technical details:

The generalized selection kernel

The collapsed kernel

Notes on Implementation

On our to do list:

- Can we characterize the efficiency gains from increasing M ?
- Can we trust the standard errors?
- $S_X(x)$ can also be computed by correctly aggregating $S(t, t')$ directly or using Poisson regression. What are the pros and cons compared to the simulated approach?

Applications

Plan for Applications

We showcase our approach with several applications using Danish administrative data.

Applications:

1. **Age differences:** Beyond binary categories—how does the matching likelihood vary continuously with age differences? A test of the “half-your-age-plus-seven” rule. Application I
2. **Two-dimensional sorting:** Joint patterns in age and education. How strong is education-based sorting *conditional* on age-based sorting? Application II
3. **Male breadwinner norm:** Are couples more likely to form/survive when men earn more? How much is driven by age-based sorting vs. gender norms? Application III
4. **Zodiac signs (placebo test):** Does “astrological compatibility” matter? Application IV

Each application demonstrates different features: continuous covariates, interactions, controls, and falsification tests.

Conclusions

Summary:

- We develop a flexible framework for measuring multidimensional marital sorting
- The (collapsed) selection kernel is the fundamental object to estimate
- Simple implementation via simulation of random couples and weighted logistic regression. Population-based approaches are available alternatives.
- Enables us to think of novel applications and revisit some old ones

Next steps:

- Formal asymptotic theory (variance, efficiency)
- Quantify simulation and approximation errors
- Show link to “workhorse” structural matching models formally. Marital surplus?
- Extensive margin (singlehood, marriage timing, divorce)
- Treatment effects: how to measure policy impacts on sorting?

Thank You

Challenges in the applied literature

How do recent applied papers take (multidimensional) sorting into account?

- Bertrand et al. (2015) construct probabilities that wives outearn their husbands by drawing from spousal-type cells (by race, age group, education, state).
- Eika et al. (2019) study education-based sorting and want to “control for” age in a robustness check. Two options: (i) condition on age (groups); (ii) construct two-dimensional sorting measure (likelihood ratios).
- Fadlon et al. (2022) use a regression-based approach—including covariates—to identify differential matching patterns across groups in their quasi-experimental setting.

Generalization: the selection kernel

We label $S(t, t')$ a *selection kernel*, defines as follows

$$S(t, t') = \frac{p(t, t')}{q(t, t')} = \frac{\text{Observed likelihood of a (t,t')-couple}}{\text{Likelihood of a (t,t')-couple under random matching}} \quad (2)$$

$q(t, t')$ can also reflect restrictions in the matching process, for example if focusing on heterosexual couples.

Generalization: the selection kernel

We label $S(t, t')$ a *selection kernel*, defines as follows

$$S(t, t') = \frac{p(t, t')}{q(t, t')} = \frac{\text{Observed likelihood of a } (t, t')\text{-couple}}{\text{Likelihood of a } (t, t')\text{-couple under random matching}} \quad (2)$$

$q(t, t')$ can also reflect restrictions in the matching process, for example if focusing on heterosexual couples.

- Computing $q(t, t')$ as the product of the marginals is computationally fast and easy for small contingency tables (low-dimensional types).
- However, this approach quickly becomes unhandy if the type space $\mathcal{T}$ is large.
 - Consider types defined as a combination of age, income, education, etc. (small bins).
 - Computational challenge to iterate through all possible combinations.
 - If t is age $\{1, 2, \dots, 100\}$, then $\mathcal{T} \times \mathcal{T}$ has already 10,000 elements

Collapsed selection kernels

In addition, often we are not interested in estimating a high dimensional $S(t, t')$ but want to consider *low-dimensions functions* defined on the type space.

Consider research questions such as:

- Are couples more likely to form if individuals have the same education?
- Are couples more likely to form if the man earns more than the wife?
- How does the likelihood of a couple forming depend on the age difference?
- How do the answers to these questions change if we control for other factors?

Answering such questions corresponds to estimating a collapsed selection kernel

$$S_X(x) = \frac{p_X(x)}{q_X(x)}, \quad (3)$$

where $x = x(t, t')$ is a vector of couple-level covariates.

Implementation

However, it is not (always) straightforward to compute $q_X(x)$ from cross-tabulations, particularly with multidimensional type spaces.

Thus, we propose estimating $S_X(x)$ using a *simulated* and *regression-based* approach:

Implementation

However, it is not (always) straightforward to compute $q_X(x)$ from cross-tabulations, particularly with multidimensional type spaces.

Thus, we propose estimating $S_X(x)$ using a *simulated* and *regression-based* approach:

1. Simulate (draw) M potential couples constructed by randomly matching individuals.
 - Stack the M potential couples and N actual couples.
 - Construct $Y = 1$ if the couple belongs to the sample of actual couples, and 0 otherwise.

Implementation

However, it is not (always) straightforward to compute $q_X(x)$ from cross-tabulations, particularly with multidimensional type spaces.

Thus, we propose estimating $S_X(x)$ using a *simulated* and *regression-based* approach:

1. Simulate (draw) M potential couples constructed by randomly matching individuals.
 - Stack the M potential couples and N actual couples.
 - Construct $Y = 1$ if the couple belongs to the sample of actual couples, and 0 otherwise.
2. Regress Y on a set of couple-level covariates capturing the type of matching in question.
 - For example, an indicator, $D = 1$ if individuals are of the same type, and 0 otherwise.
 - Weigh random/potential couples by N/M .

Implementation

However, it is not (always) straightforward to compute $q_X(x)$ from cross-tabulations, particularly with multidimensional type spaces.

Thus, we propose estimating $S_X(x)$ using a *simulated* and *regression-based* approach:

1. Simulate (draw) M potential couples constructed by randomly matching individuals.
 - Stack the M potential couples and N actual couples.
 - Construct $Y = 1$ if the couple belongs to the sample of actual couples, and 0 otherwise.
2. Regress Y on a set of couple-level covariates capturing the type of matching in question.
 - For example, an indicator, $D = 1$ if individuals are of the same type, and 0 otherwise.
 - Weigh random/potential couples by N/M .

This approach asymptotically equivalent with the standard cross-tabulation approach.

- Constructing fewer than the full set of potential couples introduces simulation bias.
- However, the bias is (likely to be) small if the dimensionality of $x(t, t')$ is small.

Application I: Age Differences

Age differences

We can go beyond simple categories like young and old. Consider how continuous age differences shape marriage market matching:

- Now we include covariates: polynomial of age differences or age-difference bins
- Estimate

$$\Pr(Y = 1|X) = \Lambda(\beta_0 + \beta' \mathbf{X})$$

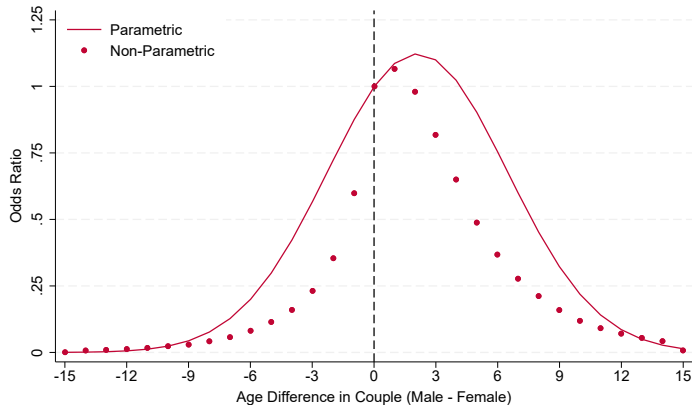
where $\mathbf{X}$ is a vector of included covariates.

Question:

1. How much more likely is it that a couple forms at different age differences relative to random matching. Are there gender differences?
2. Alternative question: do individuals follow “the rule” that you are only allowed to date someone older than half-your-age-plus-seven?

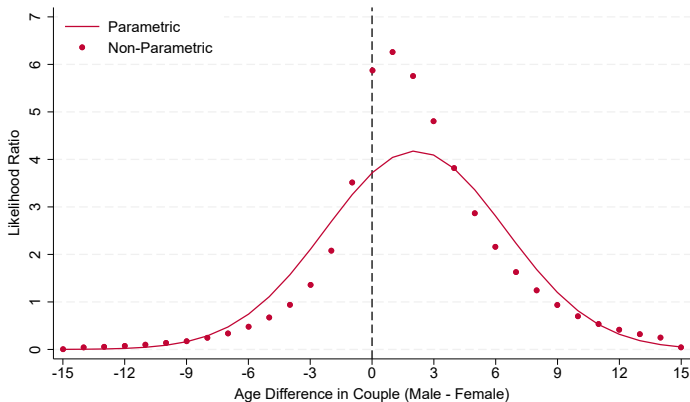
Application I: Age differences - Odds Ratios

Figure 1: The effect of age differences on the odds ratio



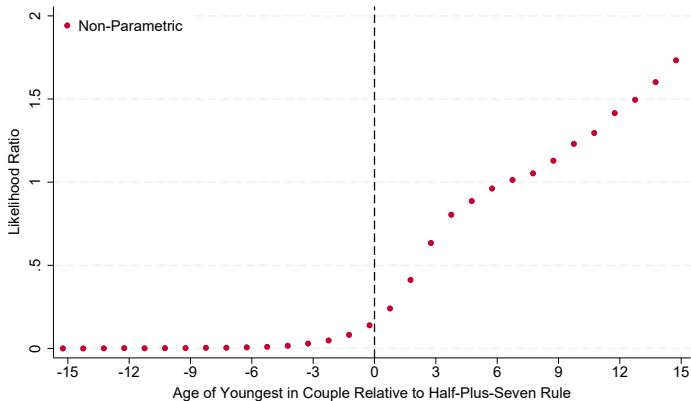
Application I: Age differences - Likelihood Ratios

Figure 2: The effect of age differences on the likelihood ratio



Application I: Age differences - “The rule”

Figure 3: The half-your-age-plus-seven rule



Application II: Two-dimensions

Age vs. Education

Application II: Two-dimensions Age vs. Education

Table 2: Distribution of Danish couples by education group in 2015

		Men					Total
		Prim.	Sec.	Voc.	BA	MA	
Women	Prim.	6.79	0.76	8.39	1.48	0.44	17.85
	Sec.	1.16	1.51	2.65	1.14	0.83	7.29
	Voc.	6.83	1.37	19.59	5.84	1.42	34.04
	BA	2.76	1.62	3.86	8.65	4.47	21.36
	MA	0.52	0.89	1.97	2.83	6.77	12.97
	Total	18.05	6.15	42.96	18.93	13.92	100.00

Application II: Two-dimensions Age vs. Education

Table 3: Distribution of Danish couples by education group in 2015

		Men					Total
		Prim.	Sec.	Voc.	BA	MA	
Women	Prim.	6.79	0.76	8.39	1.48	0.44	17.85
	<i>Expected</i>	3.22	1.10	7.67	3.38	2.48	
	Sec.	1.16	1.51	2.65	1.14	0.83	7.29
	<i>Expected</i>	1.32	0.45	3.13	1.38	1.01	
	Voc.	6.83	1.37	19.59	5.84	1.42	34.04
	<i>Expected</i>	6.14	2.09	14.62	6.44	4.74	
	BA	2.76	1.62	3.86	8.65	4.47	21.36
	<i>Expected</i>	3.86	1.31	9.18	4.04	2.97	
	MA	0.52	0.89	1.97	2.83	6.77	12.97
	<i>Expected</i>	2.34	0.80	5.57	2.45	1.81	
Total		18.05	6.15	42.96	18.93	13.92	100.00

Saturated Model: Age and Education

Specification:

Couple characteristics: $x(t, t') = (1, A, E, A \times E)$

where: $A = \mathbf{1}\{\text{same age}\}$ (Age diff < 3), $E = \mathbf{1}\{\text{same education}\}$

Regression Equation

$$\Pr(Y = 1|D) = \Lambda(\alpha + \beta_A \cdot a + \beta_E \cdot e + \beta_{AE} \cdot (a \times e))$$

Interpretation:

e^α Baseline selection (both different)

e^{β_A} Age matching gradient: $S_X(1, 0)/S_X(0, 0)$

e^{β_E} Education matching gradient: $S_X(0, 1)/S_X(0, 0)$

$e^{\beta_{AE}}$ Complementarity: $S_X(1, 1) \cdot S_X(0, 0) / S_X(1, 0) \cdot S_X(0, 1)$

Saturated Model: Age and Education - Regression Results

Variable	Education Only (1)	Age Only (2)	Both (Main Effects) (3)	Both (+ Interaction) (4)
Same Education	2.253*** (0.007)	—	2.115*** (0.007)	2.076*** (0.007)
Same Age	—	9.037*** (0.031)	8.822*** (0.031)	8.613*** (0.054)
Same Age $\times$ Same Education	—	—	—	1.080*** (0.008)
Constant	0.759*** (0.001)	0.498*** (0.001)	0.390*** (0.001)	0.392*** (0.001)
Pseudo R ²	0.026	0.163	0.181	0.181
Observations ($N = M$)	2,171,576	2,171,576	2,171,576	2,171,576

Saturated Model: Age and Education - Likelihood Ratios

Variable	Education Only (1)	Age Only (2)	Both (Main Effects) (3)	Both (+ Interaction) (4)
Same Education	1.71	—	0.82	0.81
Same Age	—	4.50	3.44	3.38
Same Age $\times$ Same Education	—	—	7.28	7.57

Saturated Model: Age and Education

Key Takeaways

- Age-based sorting is stronger than education-based sorting.
- Ignoring the interaction biases both LR_s upwards.
- Same-age-same-education couples are 7.28 times more likely to match compared to random matching, see column (3).
- The separate interaction term does not add much (+4%), see column (4).

[Back](#)

Application III: The male breadwinner norm

Application III: Male breadwinner norm and controls

Are couples more likely to form (or survive) if the man earns more than then woman?
(Bertrand et al., 2015)

- Competing explanations:
 - Gender norms / preferences
 - Mechanical sorting on age (men typically older $\rightarrow$ higher earnings) and other characteristics
- Define $D = 1$ if the male earnings $>$ female earnings, 0 otherwise.
- We estimate breadwinner effect controlling for age and location

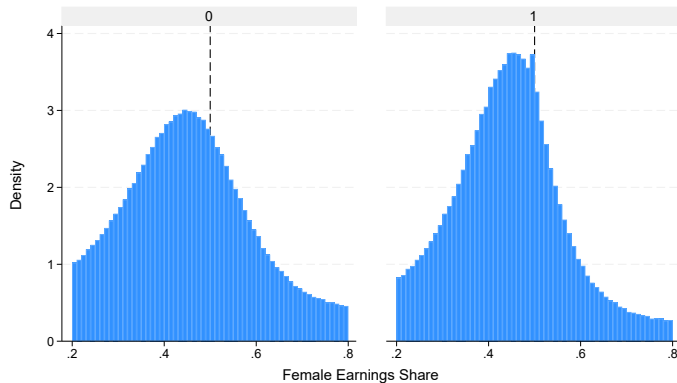
$$\Pr(Y = 1|X, D) = \Lambda(\beta_0 + \beta_1 D + \beta' \mathbf{X})$$

where $\mathbf{X}$ is a vector of other covariates:

- Age difference bins
- Place of birth (region and municipality)

Application III: Male breadwinner norm and controls

Figure 4: Random couples (left) vs. Actual couples (right)



Graphs by actual

Application III: Male breadwinner norm and controls

Table 4: Regression results

$\exp(\beta_1)$	1.302 (0.005)	1.134 (0.005)	1.105 (0.005)
$\exp(\beta_0)$	0.913 (0.003)	5.536 (0.050)	3.728 (0.036)
$N = M$	1.085.788	1.085.788	1.085.788
Age Controls	None	Diff	Diff Bins
Place of Birth Controls	None	None	Dummies

Application III: Male breadwinner norm and controls

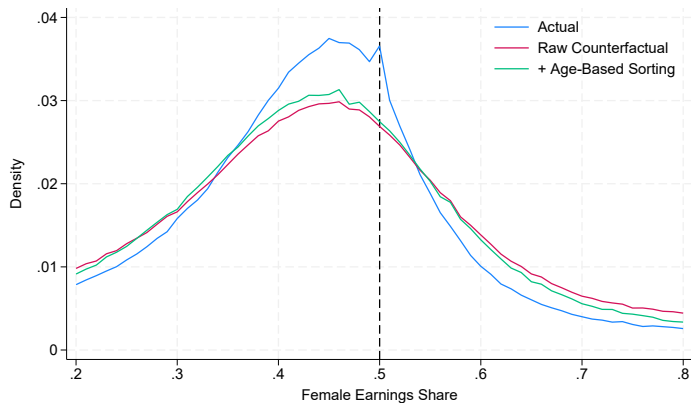
Table 4: Regression results

$\exp(\beta_1)$	1.302 (0.005)	1.134 (0.005)	1.105 (0.005)
$\exp(\beta_0)$	0.913 (0.003)	5.536 (0.050)	3.728 (0.036)
$N = M$	1.085.788	1.085.788	1.085.788
Age Controls	None	Diff	Diff Bins
Place of Birth Controls	None	None	Dummies

- Male breadwinner “norm” is to a large extent explained by age-based sorting.
- The excess odds of 30% are reduced to 13%. Place of birth matters relatively little.

Application III: Male breadwinner norm and controls

Figure 5: Random couples (left) vs. Actual couples (right)



Application IV: Zodiac Signs & Soulmates

Zodiac Signs & Soulmates

Table 5: Zodiac Soulmate Matrix (according to astrology expert Imani Quinn)

	Aries	Taurus	Gemini	Cancer	Leo	Virgo	Libra	Scorpio	Sagitta	Capric	Aqua	Pisces
Aries		.	.	.	.	.	X	X	X	.	.	.
Taurus	.		.	X	.	.	X	X	.	.	.	.
Gemini	.	.		.	.	X	.	.	X	.	.	X
Cancer	.	X	.		X	.	.	.	.	X	.	.
Leo	.	.	.	X		.	X	.	.	(X)	X	.
Virgo	.	.	X	.	.		.	X	.	.	.	X
Libra	X	X	.	.	X	.		.	.	.	.	.
Scorpio	X	X	.	.	.	X	.		.	.	.	.
Sagitta	X	.	X	.	.	.	.	.		.	.	X
Capric	.	.	.	X	(X)	.	.	.	.		X	.
Aqua	.	.	.	.	X	.	.	.	.	X		X
Pisces	.	.	X	.	.	X	.	.	X	.	X	

Zodiac Signs & Soulmates

Regression model

$$\Pr(Y = 1|X, D) = \Lambda(\beta_0 + \beta_1 D)$$

where $D = 1$ if the couple consists of zodiac soulmates.

Table 6: Regression results

$\exp(\beta_1)$	1.000 (0.003)
$\exp(\beta_0)$	1.000 (0.002)
$N = M$	1.085.788

Interpretation: No evidence that zodiac compatibility affects marriage formation.

Odds, Odds Ratios, and Likelihood Ratios

Concept	Formula	Interpretation
Odds	$\frac{P(\text{event})}{P(\text{not event})}$	Not interpretable on its own
Odds Ratio	$\frac{\text{Odds}(A)}{\text{Odds}(B)} = \exp(\beta_1)$	<i>Relative</i> comparison between groups
Likelihood Ratio	$\frac{p(t, t')}{q(t, t')}$	<i>Absolute</i> comparison to random matching

Example from age sorting:

- $LR(\text{same age}) = 1.798 \rightarrow 80\%$ more likely than random matching
- $LR(\text{diff age}) = 0.192 \rightarrow 81\%$ less likely than random matching
- $OR = 1.798/0.192 = 9.375 \rightarrow$ same-age couples have $9\times$ higher odds

Key point: OR is *relative strength*, LR is *absolute deviation* from random matching.

Why Likelihood Ratios Matter

Odds ratios alone can be misleading:

Suppose $OR = 2.0$ for same-education couples. This could mean:

- Scenario 1: $LR(\text{same}) = 2.0$, $LR(\text{diff}) = 1.0$
Same-edu over-represented, diff-edu as expected
- Scenario 2: $LR(\text{same}) = 0.4$, $LR(\text{diff}) = 0.2$
Both under-represented, but same-edu relatively less so
- Scenario 3: $LR(\text{same}) = 4.0$, $LR(\text{diff}) = 2.0$
Both over-represented, same-edu more so

Same odds ratio, completely different economic stories!

In regression: $\exp(\beta_0)$ gives the baseline LR, $\exp(\beta_1)$ gives the OR, and their product gives the LR for the treatment group.